

From Wafers to Bits and Back again: Using Deep Learning to Accelerate the Development and Characterization of SiC

R. Leonard^{1*}, M. Conrad¹, E. Van Brunt¹, J. Giles¹, E. Hutchins¹, E. Balkas¹

¹Wolfspeed, A Cree Company, 4600 Silicon Dr. Durham, NC 27703 USA

*Robert.Leonard@wolfspeed.com

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Abstract. A non-destructive, fast and accurate extended defect counting method on large diameter SiC wafers is presented. Photoluminescence (PL) signals from extended defects on 4H-SiC substrates were correlated to the specific etch features of Basal Plane Dislocations (BPDs), Threading Screw Dislocations (TSDs), and Threading Edge Dislocations (TED). For our non-destructive technique (NDT), automated defect detection was developed using modern deep convolutional neural networks (DCNN). To train a robust network, we used our large volume data set from our selective etch method of 4H-SiC substrates, already established based on definitive correlations to Synchrotron X-Ray Topography (SXRT) [1]. The defect locations, classifications and counts determined by our DCNN correlate with the subsequently etch-delineated features and counts. Once our network is sufficiently trained we will no longer need destructive methods to characterize extended defects in 4H-SiC substrates.

Introduction

The use of silicon carbide accelerates the automotive industry's transformation to electric vehicles, enabling greater system efficiencies, while reducing cost, lowering weight and conserving space [2]. Wolfspeed, as the largest volume SiC substrate producer, fuels this transformation by improving the quality and increasing the size of commercial SiC wafers [3]. Increased device yields and manufacturing efficiency require continual reduction in extended defects [4, 5]. The quick, accurate and standardized characterization of these defects is essential for crystal growth technology development and to provide reliable information to customers.

The goal of this work is to provide a non-destructive defect count method for SiC that can be implemented in the high-volume manufacturing environment. With the non-destructive method, wafers can be characterized and subsequently used for device fabrication, vastly reducing the expense of the characterization process. This advancement mainly accelerates the feedback loop for tighter process control in high volume production. Congruently, it opens additional development pathways for higher yielding device processing.

Traditionally, counting extended defects in SiC is done by delineating etch pits and counting them manually, or more recently, with automated microscopy tools. Etching reveals features that can be consistently recognized and correlated to other accepted characterization methods such as, synchrotron x-ray topography SXRT [1]. Wafer etching, however, is destructive, expensive, requires corrosive chemistries and constant attention to maintain a viable process. Furthermore, only a few wafers per crystal can be sampled, limiting the amount of information available for process improvement and control.

Photoluminescence as a non-destructive technique (PL-NDT) can be used to produce images of unetched wafers. Mapping the ultraviolet (UV) excited photoluminescence (PL) emission in a SiC wafer is useful to understand the distribution of the defects in the substrate underlying potential devices to be fabricated. PL has been established as a characterization method for extended defects in

epitaxial 4H-SiC [6], and extended to bulk characterization and wafer mapping by others [7-9]. As pointed out in the literature, the illuminated defects are near surface only, due to the small penetration depth of the UV excitation. Our effort combines etch defect information to classify the defect type and pinpoint the location in a signal obtained from PL-NDT imaging.

Recently, deep convolutional neural networks (DCNN) have dominated computer vision tasks such as image classification and object detection in a host of different contexts from cell phones to satellite images and MRI's [10-16]. Its success can be attributed to its ability to build and learn a complex combination of image filters for the specific task at hand. For a DCNN to learn to correctly classify an image or locate an object within an image, many annotated images are needed to achieve a result that generalizes features well. During the training process, DCNN receives an input image and processes it through the network of convolutional layers. The output of the DCNN is a probability distribution of classes and pixel locations. The objective of the DCNN is to minimize the error between the predicted class/location and annotated class/location.

Materials and Equipment

In this work, 150mm 4° off axis {0001} double side CMP n-type 4H-SiC wafers were characterized with a LaserTec Sica88 automatic scanning optical microscopy equipped with PL capability. PL and surface images were acquired for: 1. both Si and C face CMP wafers, and 2. post defect etching to delineate the etch pit defects for BPD, TD, and TSD's, as reported in [1]. A near infrared (NIR) long pass filter was placed before the detector to discriminate the NIR PL emission. The resulting images were processed as necessary to align features on both faces of the same wafer for characterization.

Results and Discussion

Typical images for both Si and C faces for the as polished and etched wafer are shown in Fig. 1 and Fig. 2. Fig. 1(a) shows the PL image of a portion of the measured 4H-SiC 4° off axis Si face (0001) wafer, with dark linear and concentrated dark spots in a (somewhat) noisy background. This is consistent with the work of Kawahara [9], where PL imaging produced light and dark contrast regions corresponding to dislocations in the near surface ($\sim <10\mu\text{m}$) bulk n-type wafer. The corresponding positions in the image in Fig. 1 (b) reveal the etch pit associated with the PL features. On close examination of these images, the linear features of the PL image correspond to scalloped shaped BPD etch pits and the dark spots correspond to rounded or even somewhat hexagonal etch pits, in some cases, which correspond to threading dislocations (TD), either threading edge (TED), or threading screw (TSD). From SXRT observations coupled with 4H-SiC defect etching, Sumakeris [1] reported on the inability to determine if the etch pit was of edge or screw nature just from etching the Si face; the C face etch was needed to determine which etch feature were TSD. Fig. 2, similarly, shows the PL image (a) and corresponding etch image for the C face. PL dark spots are marked with their respective etch feature. Not surprisingly, due to the difference in etching nature of the Si and C faces, not all C face dislocation etch pits correspond to the PL signal dark spots, and none of the linear features have a corresponding scalloped BPD etch pit.

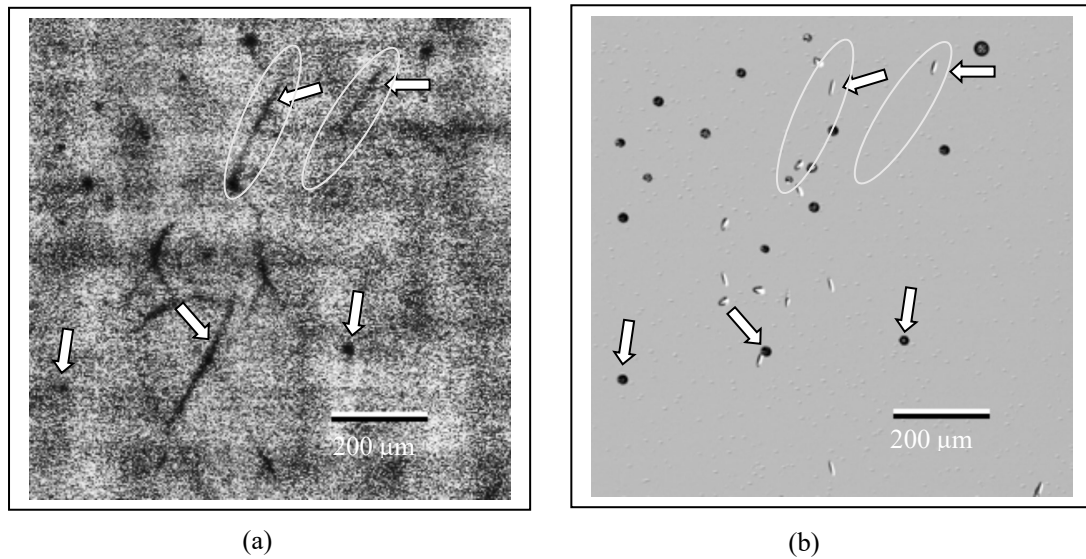


Fig. 1. Example of a Si face PL-NDT signal (a) with example corresponding etch pit image (b). Linear features within ovals in (a) correspond to scalloped shaped BPD features in (b); dark spots in (a) correspond to rounded TD features in (b). Arrows indicate corresponding defect pairs between images.

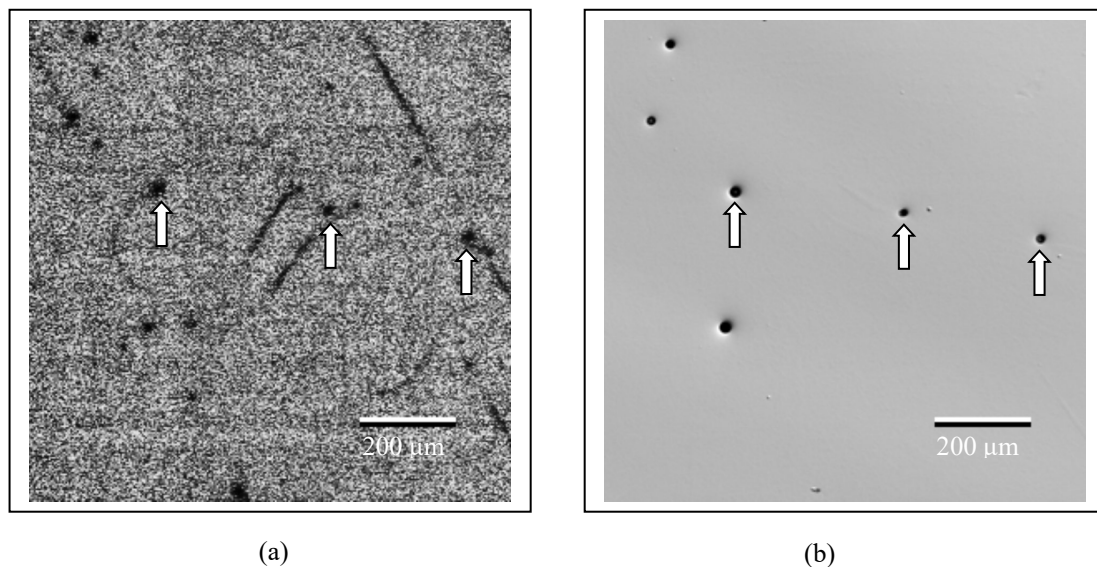


Fig. 2. Example of a C face PL-NDT signal (a) with example corresponding etch pit image (b). The PL dark spots correspond to TSD etch pits seen in (b). Arrows indicate corresponding defect pairs between images.

The examples in Fig.1 and Fig. 2 suggest that, in general, if you can see the defect signal, PL-NDT or etched, you should be able to count it. In the case of the etched feature, the contrast is straight forward to distinguish features. The etch features can be counted with commercially developed algorithms. In the PL-NDT images, however, subtle differences in contrast exist throughout the image. To count features from this nuanced signal, we need to use deep convolutional neural network (DCNN) machine learning for automated defect detection and classification.

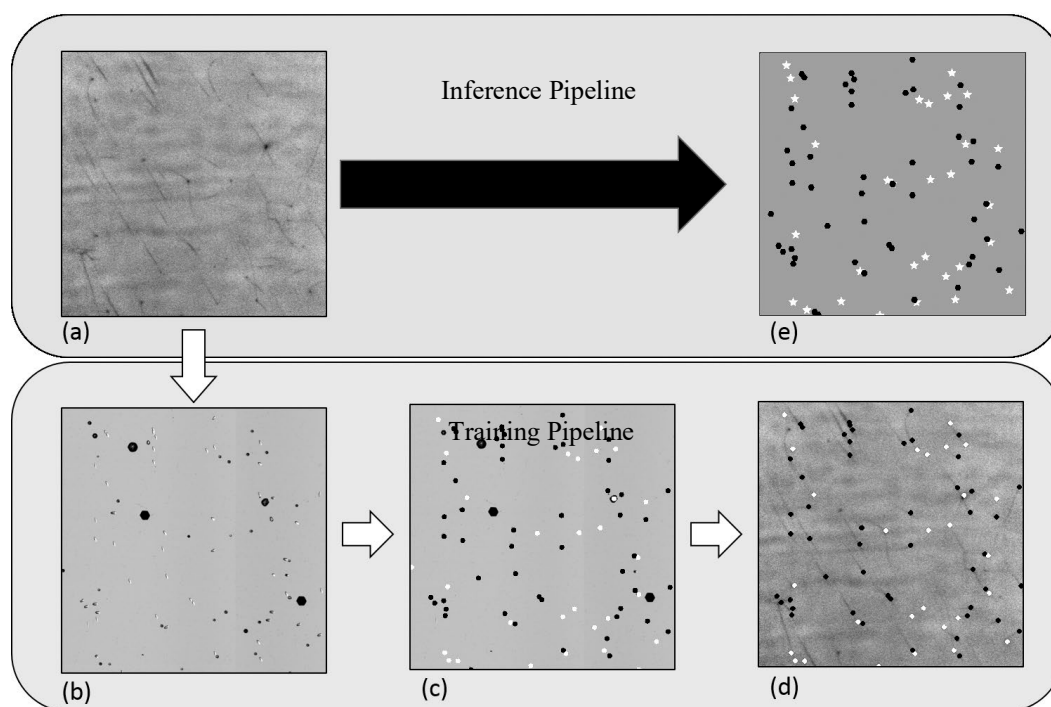


Fig. 3. a) PL-NDT optical micrograph image from Si face wafer as input image; b) micrograph of the same wafer area imaged after etching; c) defect labels generated with automated microscope inspection to create the labelled dataset for DCNN training (TD: white, BPD: black); d) etch labels transferred to PL-NDT image; e) dislocation type and location predicted by DCNN using only the PL image (TD: white stars, BPD: black dots). After training the DCNN, the black arrow indicates that only the PL-NDT measurement image is needed to produce the location and classification required for defect counting.

Fig. 3 illustrates the production of a DCNN process for non-destructive characterization of SiC. In Fig. 3 (a) a PL image of the unetched Si face wafer is shown containing features to be associated with etch defects. The *same* wafer is then etched in a eutectic etch bath of NaOH and KOH mixture to highlight the threading dislocation (TD) and basal plane dislocations (BPD). The resultant etch features (Fig 3(b)) on the wafer are imaged and labelled (Fig 3(c)) by the scanning optical microscope and software. The classified defect positions are converted to a labelling image which are spatially aligned with the initial PL image. Fig. 3 (d) shows the defect label image overlay on the PL image. A DCNN is trained with thousands of these input image/ label pairs. Automated optical microscopy equipped with PL imaging satisfies the need for large DCNN datasets. Thousands of NDT images are recorded for each wafer containing many thousands of labelled defects. The architecture used for the DCNN is consistent with the state of the art as in references [10-16], with proprietary changes made to optimize performance on our data set. Once the DCNN is trained, and using PL images from unetched wafers taken under conditions similar with those used for the training, the DCNN may be applied to infer the type and position of the defects, Fig. 3 (e). In this example, we show the conditions to detect features for the Si face of a 4H-SiC wafer. The same process is used to produce a DCNN for the TSD's on the C face, using similar images as in Fig. 2, above. The new C face DCNN was successful in classifying and locating TSD's.

With observations that the PL-NDT images of both C and Si faces are very similar, the DCNN produced for TSD's from the C face was applied to the Si Face PL-NDT images, Fig. 4. The circled dark features in Fig. 4(a) are the TSD's found in the near surface Si face PL scan determined by the DCNN. As mentioned previously, the TSD etch features on the Si face are not easily determined, since there are multiple etch behaviours associated with the TSD on the Si face. Fig. 4(b) shows the observed etch features of the Si face corresponding to the PL signal, with many threading dislocations

in the etch image, obscuring the actual position of the TSD's. Fig. 4(c), however, shows the corresponding TSD etch pit (with the x-axis reversed) aligned to the Si face revealing the same pit pattern as detected by the dark spots in the Si face PL, thus, validating the inferred result. The ability to use the C face DCNN on the Si face allows reducing the process to imaging a wafer only once on the Si face. This also allows defects to be determined on the face that is closest to the epi surface being used for devices, regardless of C or Si face. Furthermore, a limitation on the Si face etch is that both TED's and TSD's are counted at the same time. With the ability to count the TSD's on the Si face, the number of TED's may be determined as well.

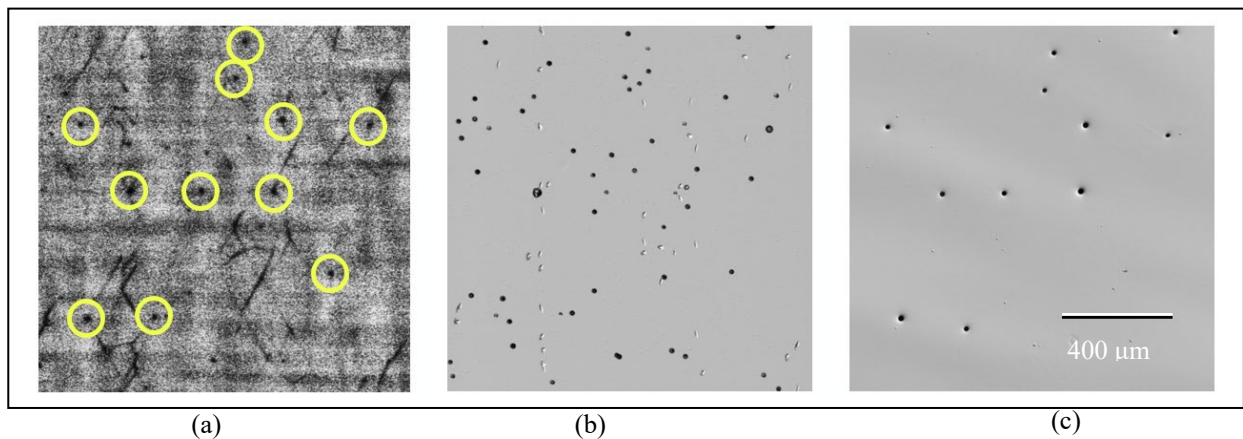


Fig. 4. (a) *Si face* 4H-SiC imaged with PL-NDT. The circles indicate the DCNN inferred position of TSD's from the PL image, not from etch labelling; (b) Defect etched Si face image showing TD and BPD etch features; (c) Defect etched C face image of same wafer area (reversed on x-axis) revealing distinct TSD etched features. All images are from same area on wafer.

Fig. 5 shows initial results as a simple validation of the PL-NDT process to match BPD densities obtained by the conventional etch process. For this data set, $N=308$ wafers were imaged with PL and subsequently processed through the DCNN network. A linear fit was applied to NDT values versus the accepted etch values with a slope of 1.13 and coefficient of determination of the near $R^2 = 0.84$. Of course, subsequent training and optimizations of the PL-NDT process will occur to improve the correlation.

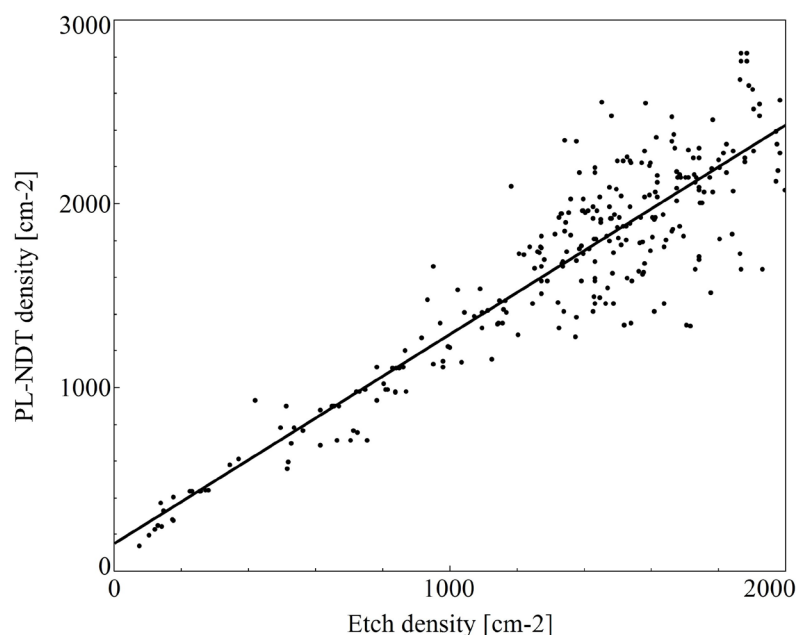


Fig. 5. BPD densities determined by PL-NDT vs. Etch process, with linear fit of data. (slope=1.13, $R^2 = 0.84$, $N=308$).

Summary

We have shown a non-destructive count method for defects in 4H-SiC is not only possible, but is close to being used for defect reporting purposes. Using non-destructive PL images of unetched wafers coupled with automatically labelled images of the corresponding etched wafers as the training set, DCNN are used to derive a network to infer the position of the defects only from the PL-NDT scan, regardless of the face scanned. The defect locations and classifications determined by DCNN correlate well with the subsequently etch delineated features, used to validate the technique. After the DCNN is sufficiently trained and validated, there is no need to destroy wafers for defect characterization, and counting by etching will be phased out. Next steps involve comparing high volume manufacturing etch defect densities to the DCNN produced densities.

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